

Empirical Proof Record: VALIDATED

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Proof ID: bc2bf41d7649918eb0e66c6bd022cac4...

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1. Claim

Across 30 synthetic three-group datasets of size 30 per group sweeping effect size from null to strong (seed=20260518), “`scipy.stats.f_oneway`“, “`statsmodels.stats.anova.anova_lm`“ (via OLS), and “`pingouin.anova`“ produce F-statistics AND two-sided p-values that agree to $\leq 1e-09$ across all pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation; three-way ANOVA variant.)

- **Operationalisation:** Generate N three-group datasets where each group has size n and is drawn from $\text{Normal}(\mu_g, \sigma^2)$ for a sweep of (μ_0, μ_1, μ_2) configurations including null (all μ equal) and alternatives. Compute F and two-sided p under `scipy.stats.f_oneway`, `statsmodels OLS+anova_lm`, and `pingouin.anova`. Compute `max_abs_diff` = max over (dataset, statistic $\in \{F, p\}$, backend-pair) of $|\text{stat}_a - \text{stat}_b|$. VALIDATED iff `max_abs_diff` $\leq$ tolerance.
- **Threshold:** `max_absolute_anova_difference` $\leq 1e-09$ F_or_p
- **H0:** At least one backend pair disagrees on the ANOVA F statistic OR the two-sided p value by more than the floating-point tolerance.
- **H1:** All three backends compute identical F AND p to floating-point tolerance across every dataset.

2. Verdict

VALIDATED — observed `7.105427357601002e-14`

30 datasets $\times$ 3 backends $\times$ 2 statistics; max pairwise $|\Delta| = 7.105e-14$ on (F, `scipy_vs_statsmodels`) ($\leq 1e-09$ tol)

3. Evidence

Pillar	Statistic	Value
anova_scipy_vs_statsmodels_vs_penguin	max_absolute_anova_difference	7.105427357601002e-

4. Pre-registration

- Registered at: 2026-05-20T15:34:38.018816+00:00
- Config hash: c8e368fe8b61be3829c27bb61d9b3208...
- Data hash: c8e368fe8b61be3829c27bb61d9b3208...

5. Signature

922d4d2cea6fddd01aa105f77aef6502d8a7e840a37ee2925f1c66ddefd9e612